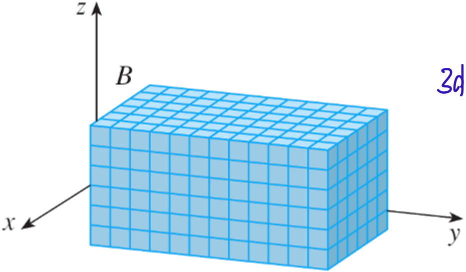
We will cover the following six topics in the Week 7 lectures:

1. Triple integrals over a closed standard box in $\mathbb{R}^3$. linked
2. Iterated triple integrals and Fubini's Theorem.
3. Triple integrals over a general bounded region in $\mathbb{R}^3$ (not necessarily closed).
4. Triple integrals over a Type-I/Type-II/Type-III region in $\mathbb{R}^3$.
5. Triple integrals in cylindrical coordinates.
6. Triple integrals in spherical coordinates.

always this order $\quad x \quad y \quad z \quad \to$ 3 closed and bounded intervals

Let $B = [a, b] \times [c, d] \times [r, s]$ be a closed standard box in $\mathbb{R}^3$.

Remark: "Standard" means that each edge of the box is parallel to a coordinate axis.



3d- box

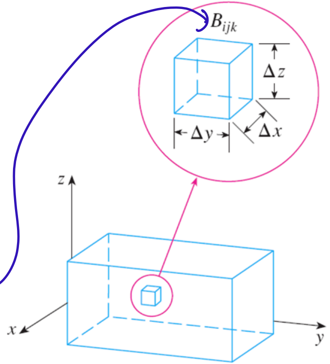
$l \cdot m \cdot n$ sub boxes

Given $l, m, n \in \mathbb{N}$, do the following:

- Divide $[a, b]$ into l bounded and closed sub-intervals of equal length, $(\Delta x)_l$.
- Divide $[c, d]$ into m bounded and closed sub-intervals of equal length, $(\Delta y)_m$.
- Divide $[r, s]$ into n bounded and closed sub-intervals of equal length, $(\Delta z)_n$.

Note that

$$(\Delta x)_l = \frac{b-a}{l}, \quad (\Delta y)_m = \frac{d-c}{m}, \quad \text{and} \quad (\Delta z)_n = \frac{s-r}{n}.$$



if l, m, n are fixed, then we don't write them down

Given $(l, m, n), (i, j, k) \in \mathbb{N}^3$ such that $1 \leq i \leq l$, $1 \leq j \leq m$, and $1 \leq k \leq n$, denote by $B_{i,j,k}^{l,m,n}$ the closed standard sub-box of B formed by taking the Cartesian product of the following three bounded and closed intervals:

- The x-side: $\left[a + (i-1)(\Delta x)_l, a + i(\Delta x)_l\right]$.
- The y-side: $\left[c + (j-1)(\Delta y)_m, c + j(\Delta y)_m\right]$.
- The z-side: $\left[r + (k-1)(\Delta z)_n, r + k(\Delta z)_n\right]$.

Not tested

A sampling of B is defined as a function $\mathbf{x}^*$ with the following two properties:

- $\mathrm{Dom}(\mathbf{x}^*) = \{((l,m,n),(i,j,k)) \in \mathbb{N}^3 \times \mathbb{N}^3 \mid 1 \le i \le l,\ 1 \le j \le m,\ 1 \le k \le n\}$.
- Given $((l,m,n),(i,j,k)) \in \mathrm{Dom}(\mathbf{x}^*)$, its output under $\mathbf{x}^*$, denoted by $\mathbf{x}^*_{(l,m,n),(i,j,k)}$ and called a "sample point", is a point in the closed standard sub-box $B^{l,m,n}_{i,j,k}$ of B.

most definitions are not tested!

Definition (Triple Integrals over a Closed Standard Box in $\mathbb{R}^3$) Not tested

- Let $B = [a,b] \times [c,d] \times [r,s]$ be a closed standard box in $\mathbb{R}^3$.
- Let f be a real-valued function defined on B (not necessarily continuous).

A triple integral of f over B is then defined as a real number L such that, ☆ regardless of the sampling $\mathbf{x}^*$ of B chosen, we have

will get same answer!

Three-dimensional Riemann Sum

$$\lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f\left(\mathbf{x}^*_{(l,m,n),(i,j,k)}\right) \cdot \underbrace{(\Delta x)_l (\Delta y)_m (\Delta z)_n}_{(\Delta V)_{l,m,n}} = L;$$

volume of a sub-box

if such a real number L exists, then it is unique, in which case we denote it by

$$\iiint_B f(x,y,z)\,\mathrm{d}V.$$

*will be tested

Theorem (Fubini's Theorem for Triple Integrals over a Closed Standard Box)

- Let $B = [a, b] \times [c, d] \times [r, s]$ be a closed standard box in $\mathbb{R}^3$.
- Let f be a continuous real-valued function defined on B.

Then $\iiint_B f(x, y, z)\, \mathrm{d}V$ exists and is equal to each of the following six iterated triple integrals:

- $\int_a^b \left[\int_c^d \left[\int_r^s f(x, y, z)\, \mathrm{d}z \right] \mathrm{d}y \right] \mathrm{d}x$ and $\int_a^b \left[\int_r^s \left[\int_c^d f(x, y, z)\, \mathrm{d}y \right] \mathrm{d}z \right] \mathrm{d}x$.
- $\int_c^d \left[\int_a^b \left[\int_r^s f(x, y, z)\, \mathrm{d}z \right] \mathrm{d}x \right] \mathrm{d}y$ and $\int_c^d \left[\int_r^s \left[\int_a^b f(x, y, z)\, \mathrm{d}x \right] \mathrm{d}z \right] \mathrm{d}y$.
- $\int_r^s \left[\int_a^b \left[\int_c^d f(x, y, z)\, \mathrm{d}y \right] \mathrm{d}x \right] \mathrm{d}z$ and $\int_r^s \left[\int_c^d \left[\int_a^b f(x, y, z)\, \mathrm{d}x \right] \mathrm{d}y \right] \mathrm{d}z$.

Example

Problem: Letting $B \overset{\text{df}}{=} [0,1] \times [-1,2] \times [0,3]$, evaluate $\iiint_B xyz^2 \, dV$.

$$\iiint_B xyz^2 \, dV = \int_0^1 \left[\int_{-1}^2 \left(\int_0^3 xyz^2 \, dz \right) dy \right] dx$$
$$= \int_0^1 \left[\int_{-1}^2 \left[\frac{xyz^3}{3} \right]_{z=0}^{z=3} dy \right] dx$$
$$= \int_0^1 \left[\int_{-1}^2 9xy \, dy \right] dx$$
$$= \int_0^1 \left[\frac{9xy^2}{2} \right]_{y=1}^{y=2} dx$$
$$= \int_0^1 \frac{27x}{2} \, dx$$
$$= \left[\frac{27x^2}{4} \right]_0^1$$
$$= \frac{27}{4}$$

Triple Integrals over a Closed Standard Box in $\mathbb{R}^3$ — Example 1 (cont'd)

Shortcut:

$$\int_0^1\left[\int_{-1}^2\left(\int_0^3 xyz^2\,dz\right)dy\right]dx$$

$$=\left(\int_0^1 x\,dx\right)\left(\int_{-1}^2 y\,dy\right)\left(\int_0^3 z^2\,dz\right)$$

$$=\frac{1}{2}\cdot\frac{3}{2}\cdot 9$$

$$=\frac{27}{4}$$

xyz^2

$\Rightarrow$ seperable expression

$(x+z)^2$

$\Rightarrow$ not seperable

Triple Integrals over a General Bounded Region in $\mathbb{R}^3$ — Definition

Definition not tested

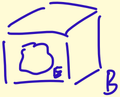
Definition (Triple Integrals over a General Bounded Region in $\mathbb{R}^3$)

- Let E be a bounded region in $\mathbb{R}^3$ (not necessarily closed). not necessarily a box
- Let f be a real-valued function defined on E (not necessarily continuous).

If B is a closed standard box in $\mathbb{R}^3$ such that $E \subseteq B$ and $\iiint_B F(x, y, z)\, \mathrm{d}V$ exists, where $F : B \to \mathbb{R}$ is defined by

B contains E

$$F(\mathbf{x}) \overset{\text{df}}{=} \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in E; \\ 0, & \text{if } \mathbf{x} \in B \setminus E, \end{cases}$$



then the triple integral of f over E, denoted by $\iiint_E f(x, y, z)\, \mathrm{d}V$, is defined as

This definition does not depend on the box chosen

$$\iiint_E f(x, y, z)\, \mathrm{d}V \overset{\text{df}}{=} \iiint_B F(x, y, z)\, \mathrm{d}V;$$

→ was defined previously

if such a closed standard box B does not exist, then we say that $\iiint_E f(x, y, z)\, \mathrm{d}V$ does not exist (or is not defined).

Definition is tested

Definition (Volume of a General Bounded Region in $\mathbb{R}^3$)

Let E be a bounded region in $\mathbb{R}^3$ (not necessarily closed).

The volume of E, denoted by $\text{Vol}(E)$, is defined as $\displaystyle\iiint_E 1 \, dV$ — if it exists.

Not every bounded region in $\mathbb{R}^3$ has a volume.

Points and line segments in $\mathbb{R}^3$ have zero volume; if

- D is a bounded and closed region in $\mathbb{R}^2$, and
- f is a continuous real-valued function defined on D,

then the graph surface of f in $\mathbb{R}^3$ also has zero volume.

2D

In fact, we have the following useful theorem.

not necessary but good to know

Not Tested(?)

Theorem (a Necessary and Sufficient Condition for the Existence of a Volume)

A bounded region in $\mathbb{R}^3$ has a volume if and only if its boundary has zero volume.

Triple Integrals over a Type-I Region in $\mathbb{R}^3$

Definitionis Tested!

Definition (Type-I Region in $\mathbb{R}^3$)

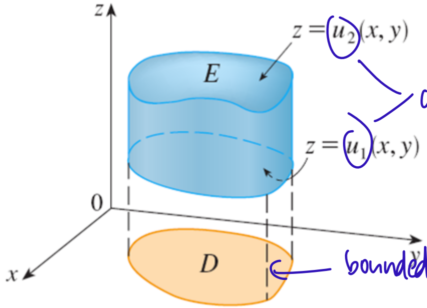
A Type I-region in $\mathbb{R}^3$ is defined as a subset E of $\mathbb{R}^3$ for which there exist

- a bounded and closed region D in $\mathbb{R}^2$, and
- continuous functions $u_1, u_2 : D \to \mathbb{R}$ satisfying $(x, y) \in D \implies u_1(x, y) \leq u_2(x, y)$,

such that

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \;\middle|\; (x, y) \in D \text{ and } u_1(x, y) \leq z \leq u_2(x, y) \right\}.$$

what we need to know



continuous

bounded and closed base

Theorem (Triple Integrals over a Type-I Region in $\mathbb{R}^3$)

- Let E continue to be the Type-I region in $\mathbb{R}^3$ from the preceding definition.
- Let f be a continuous real-valued function defined on E.

Then $\iiint_E f(x,y,z)\,\mathrm{d}V$ exists, and

outer double integral

$$\iiint_E f(x,y,z)\,\mathrm{d}V = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,\mathrm{d}z \right] \mathrm{d}A.$$

Tested!!!

Remarks:

- D is typically a Type-I or Type-II region in $\mathbb{R}^2$, so the outer double integral in the right-hand side of the equation above is typically evaluated as a Type-I or Type-II integral, respectively.
- The inner integral is an ordinary single-variable definite integral with respect to z.

Tested!

Definition (Type-II Region in $\mathbb{R}^3$)

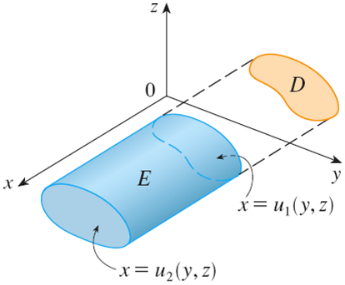
A Type II-region in $\mathbb{R}^3$ is defined as a subset E of $\mathbb{R}^3$ for which there exist

- a bounded and closed region D in $\mathbb{R}^2$, and
- continuous functions $u_1, u_2 : D \to \mathbb{R}$ satisfying $(y, z) \in D \implies u_1(y, z) \le u_2(y, z)$,

functions of y and z

such that

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \;\middle|\; (y, z) \in D \text{ and } u_1(y, z) \le x \le u_2(y, z) \right\}.$$



Theorem (Triple Integrals over a Type-II Region in $\mathbb{R}^3$)

- Let E continue to be the Type-II region in $\mathbb{R}^3$ from the preceding definition.
- Let f be a continuous real-valued function defined on E.

Then $\iiint_E f(x,y,z)\,\mathrm{d}V$ exists, and

outer double integral

$$\iiint_E f(x,y,z)\,\mathrm{d}V = \iint_D \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x,y,z)\,\mathrm{d}x \right] \mathrm{d}A.$$

Tested!

Remarks:

- D is typically a Type-I or Type-II region in $\mathbb{R}^2$, so the outer double integral in the right-hand side of the equation above is typically evaluated as a Type-I or Type-II integral, respectively.
- The inner integral is an ordinary single-variable definite integral with respect to x.

Definition (Type-III Region in $\mathbb{R}^3$)

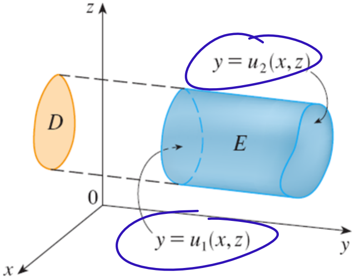
A Type III-region in $\mathbb{R}^3$ is defined as a subset E of $\mathbb{R}^3$ for which there exist

- a bounded and closed region D in $\mathbb{R}^2$, and
- continuous functions $u_1, u_2 : D \to \mathbb{R}$ satisfying $(x, z) \in D \implies u_1(x, z) \leq u_2(x, z)$,

such that

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \;\middle|\; (x, z) \in D \text{ and } u_1(x, z) \leq y \leq u_2(x, z) \right\}.$$

y is sandwiched between the graphs of 2 x-z functions



Theorem (Triple Integrals over a Type-III Region in $\mathbb{R}^3$)

- Let E continue to be the Type-III region in $\mathbb{R}^3$ from the preceding definition.
- Let f be a continuous real-valued function defined on E.

Then $\iiint_E f(x, y, z)\, \mathrm{d}V$ exists, and

outer double integral

$$\iiint_E f(x, y, z)\, \mathrm{d}V = \iint_D \left[\int_{u_1(x,z)}^{u_2(x,z)} f(x, y, z)\, \mathrm{d}y \right] \mathrm{d}A.$$

Tested!

Remarks:

- D is typically a Type-I or Type-II region in $\mathbb{R}^2$, so the outer double integral in the right-hand side of the equation above is typically evaluated as a Type-I or Type-II integral, respectively.
- The inner integral is an ordinary single-variable definite integral with respect to y.

Triple Integrals over a Type-I/II/III Region in $\mathbb{R}^3$ — Example 1

Example

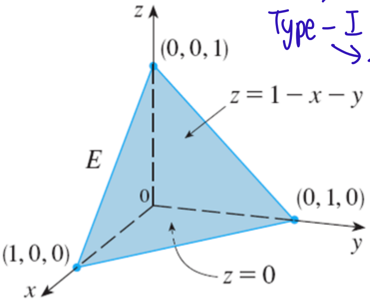
Let T denote the closed tetrahedron in $\mathbb{R}^3$ bounded by the three coordinate planes and the plane given by $x + y + z = 1$.

Problem: Evaluate $\displaystyle\iiint_T z \, \mathrm{d}V$.

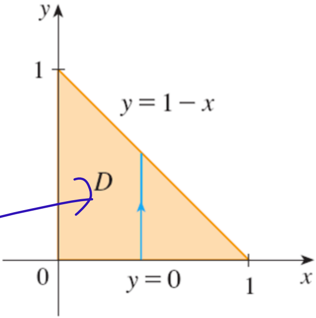
Solution

Type – I → z is sandwiched between 2 graphs

→ triple integral over z



Type I base



$$\iiint_T z\,dV = \iint_D \left[\int_0^{1-x-y} z\,dz\right] dA_{(x,y)}$$
$$= \iint_D \left[\frac{z^2}{2}\right]_0^{1-x-y} dA_{(x,y)}$$
$$= \iint_D \frac{1}{2}(1+x^2+y^2-2x-2y+2xy)\,dA_{(x,y)}$$
$$= \int_0^1 \left[\int_0^{1-x} \frac{1}{2}(1+x^2+y^2-2x-2y+2xy)\,dy\right] dx$$

Type-I

$$= \dots$$
$$= \frac{1}{2}\int_0^1 \left[-\frac{1}{3}(x-1)^3\right] dx$$
$$= \dots$$
$$= \frac{1}{24}$$

Type-I Triple Integral

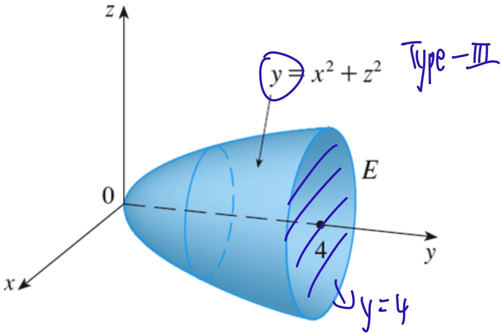
Triple Integrals over a Type-I/II/III Region in $\mathbb{R}^3$ — Example 1 (cont'd)

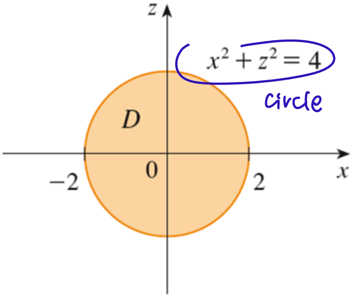
Example

Let E denote the closed region in $\mathbb{R}^3$ bounded by the paraboloid given by $y = x^2 + z^2$ and the plane given by $y = 4$.

Problem: Evaluate $\iiint_E \sqrt{x^2 + z^2}\, dV$.

Solution





$$\iiint_E \sqrt{x^2+z^2}\, dV = \iint_D \left[\int_{x^2+z^2}^{4} \sqrt{x^2+z^2}\, dy \right] dA_{(x,z)}$$

$$= \iint_D \left[\sqrt{x^2+z^2} \cdot y \right]_{y=x^2+z^2}^{y=4} dA_{(x,z)}$$

$$= \iint_D \left[4\sqrt{x^2+z^2} - (x^2+z^2)^{3/2} \right] dA_{(x,z)}$$

x-z plane
$x = r\cos\theta$
$z = r\sin\theta$

circle

can write in polar coordinates

angle radius

$$= \int_0^{2\pi} \left[\int_0^2 (4r - r^3) \cdot r\, dr \right] d\theta$$

$$= \left(\int_0^2 (4r^2 - r^4)\, dr \right) \left(\int_0^{2\pi} 1\, d\theta \right)$$

$$= \frac{64}{15} \cdot 2\pi$$

$$= \frac{128}{15}\pi$$

Triple Integrals over a Type-I/II/III Region in $\mathbb{R}^3$ — Example 2 (cont'd)

General Formula for Volume: $V = \iiint_E 1\,dV$

E is any bounded region in $\mathbb{R}^3$

Special case when E is a Type I, Type II, or Type III region in $\mathbb{R}^3$.

① Type I: $V = \iiint_E 1\,dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} 1\,dz \right] dA_{(x,y)}$ } D is in the xy-plane

lower surface: $z = u_1(x,y)$, upper surface: $z = u_2(x,y)$

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② Type II: $V = \iiint_E 1\,dV = \iint_D \left[\int_{u_1(y,z)}^{u_2(y,z)} 1\,dx \right] dA_{(y,z)}$ } D is in the yz-plane

left surface: $x = u_1(y,z)$, right surface: $x = u_2(y,z)$

---

③ Type III: $V = \iiint_E 1\,dV = \iint_D \left[\int_{u_1(x,z)}^{u_2(x,z)} 1\,dy \right] dA_{(x,z)}$ } D is in the xz-plane

Front surface: $y = u_1(x,z)$, back surface: $y = u_2(x,z)$

Triple Integrals over a Type-I/II/III Region in $\mathbb{R}^3$ — Example 3

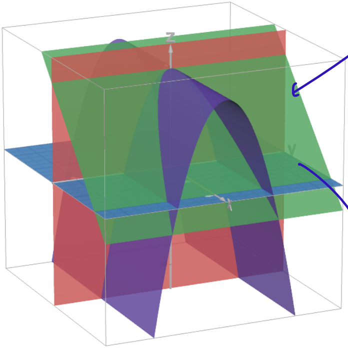
Example

Let E denote the closed region in $\mathbb{R}^3$ bounded by the surface given by $\underline{z = 4 - y^2}$ and the three planes given by $x = 0$, $z = 0$, and $\underline{x + z = 4}$.
Problem: Find Vol(E).

parabola

left surface

Solution



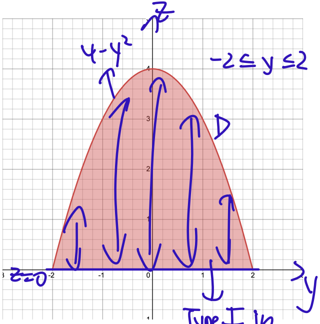
Type II region

$x = 4 - z$

$\Updownarrow$

$x + z = 4$

in the right surface



Type I in yz-plane

Triple Integrals over a Type-I/II/III Region in $\mathbb{R}^3$ — Example 3 (cont'd)

$$
\begin{aligned}
\text{Vol}(E) &= \iiint_E 1\, dV \\
&= \iint_D \left[\int_0^{4-z} 1\, dx \right] dA(y,z) \\
&= \iint_D \left[x \right]_0^{4-z} dA(x,z) \\
&= \iint_D (4-z)\, dA(y,z) \\
&= \int_{-2}^{2} \left[\int_0^{4-y^2} (4-z)\, dz \right] dy \\
&= \int_{-2}^{2} \left[4z - \frac{z^2}{2} \right]_0^{4-y^2} dy \\
&= \int_{-2}^{2} \left[4(4-y^2) - \frac{(4-y^2)^2}{2} \right] dy \\
&= \int_{-2}^{2} \left(8 - \frac{y^4}{2} \right) dy \\
&= \left[8y - \frac{y^5}{10} \right]_{-2}^{2} \\
&= \frac{128}{5} \\
&= 25.6
\end{aligned}
$$

Cylindrical Coordinates

Cylindrical coordinates, like Cartesian coordinates, provide a means of specifying points in three-dimensional space.

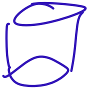
(x, y, z)

same

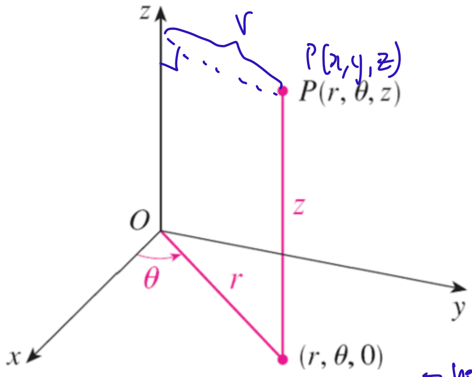
Cylindrical coordinates blend polar coordinates with the z-coordinate to specify points in $\mathbb{R}^3$. (r, θ, z)

(r, θ)

They are useful for evaluating triple integrals over regions in $\mathbb{R}^3$ that have a convenient description in terms of cylindrical coordinates, such as solid cones and solid cylinders



Cylindrical Coordinates



not from the origin!

Given a point $P = (x, y, z) \in \mathbb{R}^3$ not on the z-axis, define r and θ as follows:

- $r \in \mathbb{R}_{>0}$ is defined as the distance of P from the z-axis, equal to $\sqrt{x^2 + y^2}$.
  Remark: r is known as the radial distance of P from the z-axis. ⇒ if same x, y values but different z, r is same for both points
- $\theta \in [0, 2\pi[$ is defined as the counterclockwise angle that the line segment between the points $(0, 0)$ and (x, y) in $\mathbb{R}^2$ makes with the positive x-axis.
  Remark: θ is known as the azimuthal angle of P or just the azimuth of P.

Conversion from cylindrical coordinates to Cartesian coordinates: Straight forward

$$\underbrace{(r,\theta,z)}_{\text{in } \mathbb{R}_{\geq 0}\times\mathbb{R}\times\mathbb{R}} \mapsto (x,y,z) = (r\cos(\theta), r\sin(\theta), z).$$

dist cannot be negative

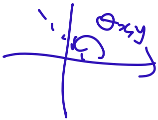
Conversion from Cartesian coordinates to cylindrical coordinates (some restrictions):

same

$$\underbrace{(x,y,z)}_{\text{in } \mathbb{R}^3 \text{ except the } z\text{-axis}} \mapsto (r,\theta,z) = \left(\underbrace{\sqrt{x^2+y^2}}_{r}, \Theta_{x,y}, z\right).$$

Remark: $\Theta_{x,y}$ denotes the unique angle $\theta \in [0, 2\pi[$ such that

$$\underbrace{\left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}}\right)}_{\text{on the unit circle}} = (\cos(\theta), \sin(\theta)).$$



$\Theta_{x,y}$ measures the counter-clockwise angle of (x,y) from positive x-axis

Example

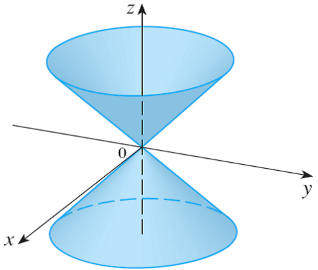
Describe the surface in $\mathbb{R}^3$ whose equation in terms of cylindrical coordinates is $|z| = r$.

$\sqrt{x^2+y^2}$

Solution

- Observe that $|z| = r \iff |z| = \sqrt{x^2+y^2} \iff z^2 = x^2 + y^2$. $\Rightarrow$ cone in $\mathbb{R}^3$
- Therefore, the surface is a cone in $\mathbb{R}^3$ with the following two properties:
  - Its rotational axis of symmetry is the z-axis.
  - Its slope is 1.

see textbook for equations of well-known surfaces in $\mathbb{R}^3$.

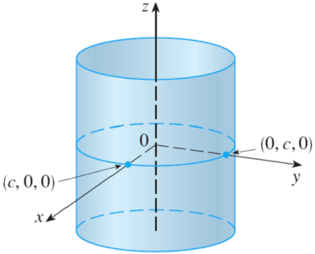


Example

Describe the surface in $\mathbb{R}^3$ whose equation in terms of cylindrical coordinates is $r = c$, where c is a positive real number.

Solution

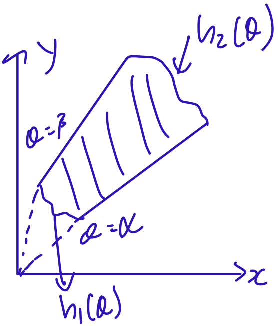
- Observe that $r = c \iff \overbrace{\sqrt{x^2 + y^2}}^{r} = c.$ dist from z-axis
- Therefore, the surface is a cylinder in $\mathbb{R}^3$ with the following two properties:
  - Its rotational axis of symmetry is the z-axis.
  - Its radius is c.

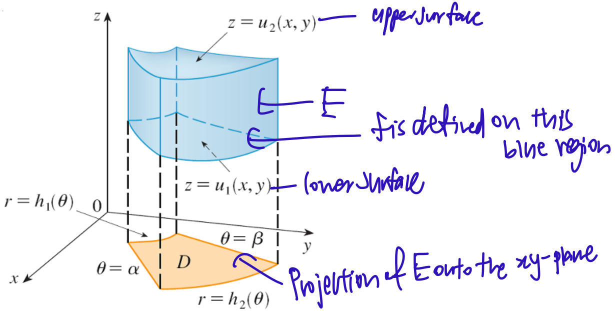


If $c=0 \Rightarrow$ get z-axis

a line

Triple Integrals in Cylindrical Coordinates





Cylindrical coordinates are very useful for evaluating triple integrals over closed regions in $\mathbb{R}^3$ whose projection onto the *xy*-plane is a polar region.

Tested!!

Theorem (Triple Integrals in Cylindrical Coordinates)

Set up for the problem

- Let $\alpha, \beta \in \mathbb{R}$ satisfy $0 \leq \beta - \alpha \leq 2\pi$. α, β – initial and final angles of polar region D.
- Let $h_1, h_2 : [\alpha, \beta] \to \mathbb{R}_{\geq 0}$ satisfy the following three properties:
  - h_1 and h_2 are continuous. → always the case
  - For all $\theta \in [\alpha, \beta]$, we have $h_1(\theta) \leq h_2(\theta)$. h_1 – inner radius, h_2 – outer radius
  - If $\beta - \alpha = 2\pi$, then $h_1(\alpha) = h_1(\beta)$ and $h_2(\alpha) = h_2(\beta)$. to ensure endpoints match after 1 cycle
- Let $D \stackrel{\text{df}}{=} \left\{(r\cos(\theta), r\sin(\theta)) \in \mathbb{R}^2 \;\middle|\; (r, \theta) \in [h_1(\theta), h_2(\theta)] \times [\alpha, \beta]\right\}$.
- Let u_1 and u_2 be continuous real-valued functions defined on D with the property that $(x, y) \in D \implies u_1(x, y) \leq u_2(x, y)$. u_1 – lower surface, u_2 – upper surface
- Finally, let f be a continuous real-valued function defined on

$$E \stackrel{\text{df}}{=} \left\{(x, y, z) \in \mathbb{R}^3 \;\middle|\; (x, y) \in D \text{ and } u_1(x, y) \leq z \leq u_2(x, y)\right\}.$$

Then $\iiint_E f(x, y, z)\, \mathrm{d}V$ exists and is equal to

Focus on this!

$$\int_{\alpha}^{\beta} \left[\int_{h_1(\theta)}^{h_2(\theta)} \left[\int_{u_1(r\cos(\theta), r\sin(\theta))}^{u_2(r\cos(\theta), r\sin(\theta))} f(r\cos(\theta), r\sin(\theta), z) \cdot r \, \mathrm{d}z \right] \mathrm{d}r \right] \mathrm{d}\theta.$$

Don't forget!!

Example

Let E denote the closed region in $\mathbb{R}^3$ that lies

- within the cylinder given by $x^2 + y^2 = 1$;
- below the plane given by $z = 4$; and
- above the paraboloid given by $z = 1 - x^2 - y^2$.

change to polar

$1-x^2-y^2 = 1-(x^2+y^2)$

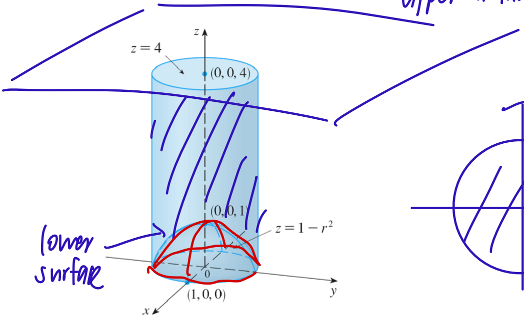
$= 1-r^2$

Problem: Evaluate $\iiint_E \sqrt{x^2+y^2}\, dV$.

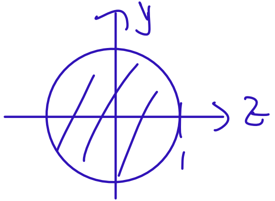
Solution

upper surface

$z=4$



lower surface



$$\iiint_E \sqrt{x^2+y^2}\, dV = \int_0^{2\pi} \left[\int_{1-r^2}^{4} \sqrt{r^2} \cdot r\, dz \right] dr \Big] d\theta$$

$$= \int_0^{2\pi} \left[\int_0^1 \left[\int_{1-r^2}^{4} r^2\, dz \right] dr \right] d\theta$$

$$= \int_0^{2\pi} \left[\int_0^1 \left[r^2 z \right]_{z=1-r^2}^{z=4} dr \right] d\theta$$

$$= \int_0^{2\pi} \left[\int_0^1 \left[4r^2 - r^2(1-r^2) \right] dr \right] d\theta$$

$$= \int_0^{2\pi} \left[\int_0^1 (r^4 + 3r^2)\, dr \right] d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^5}{5} + r^3 \right]_0^1 d\theta$$

$$= \int_0^{2\pi} \frac{6}{5}\, d\theta$$

$$= \left[\frac{6}{5}\theta \right]_0^{2\pi}$$

$$= \frac{12\pi}{5}$$

Triple Integrals in Cylindrical Coordinates — Example 2

Example

Problem: Evaluate the iterated triple integral

Change to polar coordinates!

$$\int_{-2}^{2}\left[\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\left[\int_{\sqrt{x^2+y^2}}^{2}\left(x^2+y^2\right)\,dz\right]dy\right]dx.$$

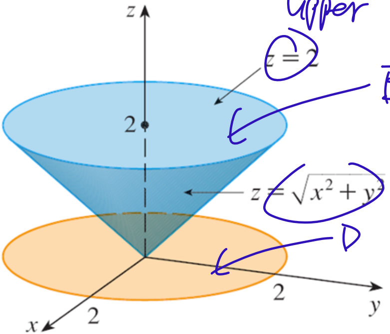
Solution

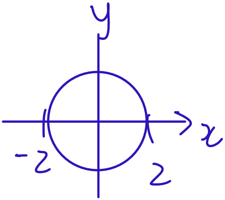
Upper

E

lower

D





remember!!

$$\int_0^{2\pi}\left[\int_0^2\left[\int_r^2 r^2\cdot r\,dz\right]dr\right]d\theta$$

$$=\int_0^{2\pi}\left[\int_0^2\left[\int_r^2 r^3\,dz\right]dr\right]d\theta$$

$$=\int_0^{2\pi}\left[\int_0^2\left[r^3 z\right]_{z=r}^{z=2}dr\right]d\theta$$

$$=\int_0^{2\pi}\left[\int_0^2(2r^3-r^4)\,dr\right]d\theta$$

$$=\left(\int_0^2(2r^3-r^4)\,dr\right)\left(\int_0^{2\pi}1\,d\theta\right)$$

$$=\left[\frac{r^4}{2}-\frac{r^5}{5}\right]_0^2\left[\theta\right]_0^{2\pi}$$

$$=\dots$$

$$=\frac{8}{5}\cdot 2\pi$$

$$=\frac{16}{5}\pi$$

Spherical Coordinates

Spherical coordinates offer yet another means of specifying points in three-dimensional space.

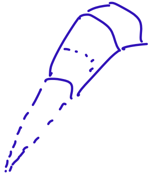
distance

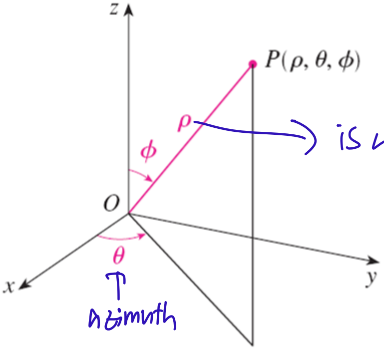
Spherical coordinates use one spatial coordinate and two angular coordinates to specify points in $\mathbb{R}^3$, while cylindrical coordinates use two spatial coordinates and one angular coordinate to do the same.

r, z

θ

They are useful for evaluating triple integrals over regions in $\mathbb{R}^3$ that have a convenient description in terms of spherical coordinates, such as solid cones and spherical wedges.





is now distance from origin

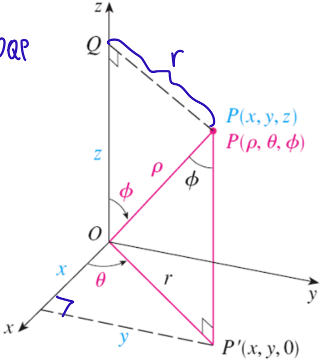
azimuth

Given a point $P = (x, y, z) \in \mathbb{R}^3$ not on the z-axis, define ρ, θ, and ϕ as follows:

- $\rho \in \mathbb{R}_{>0}$ is defined as the length of OP, equal to $\sqrt{x^2 + y^2 + z^2}$.
  Remark: ρ is known as the radial distance of P from the origin.
- $\theta \in [0, 2\pi[$ is defined as the azimuth of P.
- $\phi \in]0, \pi[$ is defined as the angle between OP and the positive z-axis.
  Remark: ϕ is known as the inclination of P or the polar angle of P.

Spherical Coordinates

z is adjacent to ϕ in ΔOQP



r is opp. of ϕ in $\underline{\Delta OQP}$
$\Downarrow$
hypotenuse OP has length ρ

Observe the following:

- $x = r\cos(\theta) = \rho\sin(\phi)\cdot\cos(\theta) = \rho\sin(\phi)\cos(\theta)$.
- $y = r\sin(\theta) = \rho\sin(\phi)\cdot\sin(\theta) = \rho\sin(\phi)\sin(\theta)$.
- $z = \rho\cos(\phi)$.

Tested

$\frac{z}{\rho} = \cos(\phi) \Leftrightarrow \frac{z}{\sqrt{x^2+y^2+z^2}} = \cos(\phi)$

Conversion from spherical coordinates to Cartesian coordinates: Straight forward

$$\underbrace{(\rho, \theta, \phi)}_{\text{in } \mathbb{R}_{\geq 0} \times \mathbb{R} \times [0,\pi]} \mapsto (x, y, z) = (\underbrace{\rho \sin(\phi) \cos(\theta)}_{x}, \underbrace{\rho \sin(\phi) \sin(\theta)}_{y}, \underbrace{\rho \cos(\phi)}_{z}).$$

Conversion from Cartesian coordinates to spherical coordinates (some restrictions):
not so straightforward.

$$\underbrace{(x, y, z)}_{\text{in } \mathbb{R}^3 \text{ except the } z\text{-axis}} \mapsto (\rho, \theta, \phi) = \left(\sqrt{x^2 + y^2 + z^2}, \underbrace{\Theta_{x,y}}_{\text{same as before}}, \arccos\left(\frac{z}{\underbrace{\sqrt{x^2 + y^2 + z^2}}_{\rho}} \right) \right).$$

Remark: $\Theta_{x,y}$ denotes the unique angle $\theta \in [0, 2\pi[$ such that

$$\left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}} \right) = (\cos(\theta), \sin(\theta)).$$

Example

Describe the surface in $\mathbb{R}^3$ whose equation in terms of spherical coordinates is $\rho = c$, where c is a positive real number.

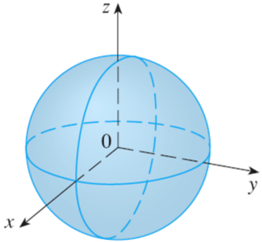
dist from origin is fixed

$\therefore$ sphere

Solution

- Observe that $\rho = c \iff \sqrt{x^2 + y^2 + z^2} = c$.
- Therefore, the surface is a sphere in $\mathbb{R}^3$ with center $(0, 0, 0)$ and with radius c.

$x^2 + y^2 + z^2 = c^2$



Example

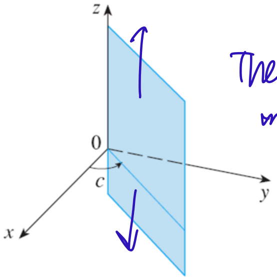
Describe the surface in $\mathbb{R}^3$ whose equation in terms of spherical coordinates is $\theta = c$.

azimuth

Sensitive to the quadrant you are in.

Solution

- The surface is a closed vertical half-plane in $\mathbb{R}^3$ with the following two properties:
  - Its boundary is the z-axis.
  - Its points off of the z-axis all have an azimuth of c.



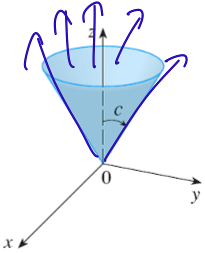
The plane extend upwards and downward.

Example

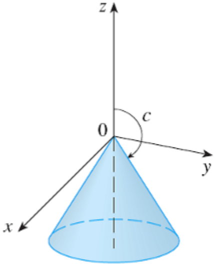
Describe the surface in $\mathbb{R}^3$ whose equation in terms of spherical coordinates is $\phi = c$, where $c \in]0, \pi[$.

Solution

- The surface is a half-cone in $\mathbb{R}^3$ with the following three properties:
  - Its rotational axis of symmetry is the z-axis.
  - Its vertex is $(0, 0, 0)$.
  - Its points except for the vertex all have a polar angle of c.



$0 < c < \pi/2$



$\pi/2 < c < \pi$

$c = \frac{\pi}{2}$
$\Rightarrow$ xy plane
$\phi = 0$
$\Rightarrow$ positive z-axis
$\phi = \pi$
$\Rightarrow$ negative z-axis

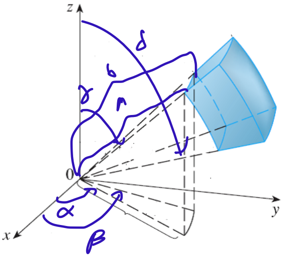
Triple Integrals in Spherical Coordinates — Nice Spherical Wedges

A nice spherical wedge is a bounded and closed region E in $\mathbb{R}^3$ for which there exist

- $a, b \in \mathbb{R}_{\geq 0}$ satisfying $a \leq b$, a - inner radius, b - outer radius
- $\alpha, \beta \in \mathbb{R}$ satisfying $0 \leq \beta - \alpha \leq 2\pi$, and α - initial azimuth, β - final azimuth
- $\gamma, \delta \in [0, \pi]$ satisfying $\gamma \leq \delta$, γ - initial polar angle, δ - final polar angle

such that E is specified by

$$\left\{ \left(\underbrace{\rho \sin(\phi)\cos(\theta)}_{x}, \underbrace{\rho \sin(\phi)\sin(\theta)}_{y}, \underbrace{\rho\cos(\phi)}_{z} \right) \in \mathbb{R}^3 \;\middle|\; (\rho, \theta, \phi) \in [a, b] \times [\alpha, \beta] \times [\gamma, \delta] \right\}.$$



In spherical coordinates, a nice spherical wedge is represented as a closed standard box.

Theorem (Triple Integrals in Spherical Coordinates for Nice Spherical Wedges)

- Let E be the nice spherical wedge specified on the preceding slide.
- Let f be a continuous real-valued function defined on E.

Then $\iiint_E f(x, y, z)\, \mathrm{d}V$ exists and is equal to

Tested.

$$\int_\gamma^\delta \left[\int_\alpha^\beta \left[\int_a^b f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \cdot \rho^2 \sin(\phi)\, \mathrm{d}\rho \right] \mathrm{d}\theta \right] \mathrm{d}\phi.$$

The factor of $\rho^2 \sin(\phi)$ is known as the Jacobian of the coordinate transformation from spherical coordinates to Cartesian coordinates.

Jacobians will be formally defined when we learn the Change-of-Variables Formula.

They are deformation factors that are incurred when we switch coordinate systems.

Tested

Theorem (Triple Integrals in Spherical Coordinates for Type-II Spherical Wedges)

- Let $\alpha, \beta \in \mathbb{R}$ satisfy $0 \leq \beta - \alpha \leq 2\pi$, and let $\gamma, \delta \in [0, \pi]$ satisfy $\gamma \leq \delta$.
- Let $g_1, g_2 : [\alpha, \beta] \times [\gamma, \delta] \to \mathbb{R}_{\geq 0}$ satisfy the following three properties:
  - g_1 and g_2 are continuous.
  - Given $(\theta, \phi) \in [\alpha, \beta] \times [\gamma, \delta]$, we have $g_1(\theta, \phi) \leq g_2(\theta, \phi)$.
  - If $\beta - \alpha = 2\pi$, then $(\forall \phi \in [\gamma, \delta])(g_1(\alpha, \phi) = g_1(\beta, \phi)$ and $g_2(\alpha, \phi) = g_2(\beta, \phi))$.

ensures endpoint match after one full cycle

- Finally, let f be a continuous real-valued function defined on

$$E \stackrel{\text{df}}{=} \left\{ (\underbrace{\rho \sin(\phi) \cos(\theta)}_{x}, \underbrace{\rho \sin(\phi) \sin(\theta)}_{y}, \underbrace{\rho \cos(\phi)}_{z}) \in \mathbb{R}^3 \;\middle|\; \begin{array}{l} (\theta, \phi) \in [\alpha, \beta] \times [\gamma, \delta], \\ \rho \in [g_1(\theta, \phi), g_2(\theta, \phi)] \end{array} \right\}.$$

inner radius func. outer radius func.

Focus on this

Then $\iiint_E f(x, y, z)\, \mathrm{d}V$ exists and is equal to

$$\boxed{\int_\gamma^\delta \left[\int_\alpha^\beta \left[\int_{g_1(\theta,\phi)}^{g_2(\theta,\phi)} f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \cdot \rho^2 \sin(\phi)\, \mathrm{d}\rho \right] \mathrm{d}\theta \right] \mathrm{d}\phi.}$$

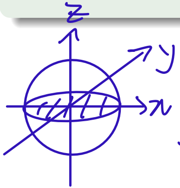
Remark: In spherical coordinates, E is represented as a Type-II region.

Similar to previous slide

Example

Let $B \overset{\text{df}}{=} \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1\}$ — the closed unit ball in $\mathbb{R}^3$.

Problem: Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{\frac{3}{2}}}\, \mathrm{d}V$.



B in spherical coordinates
$$\begin{cases} 0 \le \rho \le 1 \\ 0 \le \theta \le 2\pi \\ 0 \le \phi \le \pi \end{cases}$$

(also can $-\pi \le \theta \le \pi$) as long as make one full cycle, can

$$\int_0^{\pi} \left[\int_0^{2\pi} \left[\int_0^1 e^{(\rho^2)^{\frac{3}{2}}} \cdot \rho^2 \sin(\phi)\, d\rho \right] d\theta \right] d\phi$$

seperable expression

$$= \int_0^{\pi} \left[\int_0^{2\pi} \left[\int_0^1 \rho^2 e^{\rho^3} \sin(\phi)\, d\rho \right] d\theta \right] d\phi$$

$$= \left(\int_0^1 \rho^2 e^{\rho^3} d\rho\right)\left(\int_0^{2\pi} 1\, d\theta\right)\left(\int_0^{\pi} \sin(\phi)\, d\phi\right)$$

$$= \left[\frac{1}{3}e^{\rho^3}\right]_0^1 \left[\theta\right]_0^{2\pi} \left[-\cos(\phi)\right]_0^{\pi}$$

$$= \frac{1}{3}(e-1) \cdot 2\pi \cdot 2$$

$$= \frac{4\pi}{3}(e-1)$$

Difficult Problem — Pay ATTENTION!!!

Example

Let E denote the closed region in $\mathbb{R}^3$ that lies

- above the cone given by $z = \sqrt{x^2 + y^2}$; and
- below the sphere given by $x^2 + y^2 + z^2 = z$.

cone has slope 1

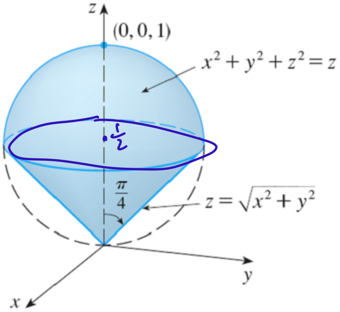
E $x^2 + y^2 + z^2 - z = 0$

$x^2 + y^2 + (z - \frac{1}{2})^2 = (\frac{1}{2})^2$

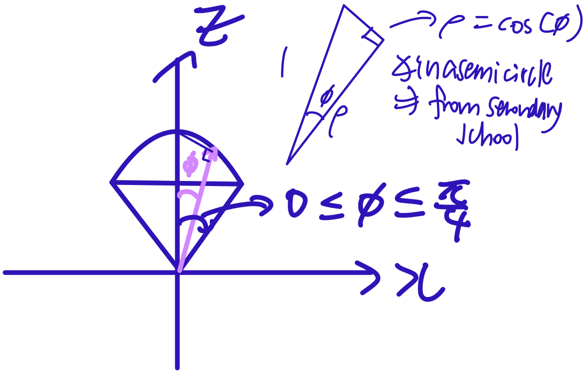
sphere with centre $(0, 0, \frac{1}{2})$, radius : $\frac{1}{2}$

Problem: Find Vol(E).

Solution



$0 \le \theta \le 2\pi \qquad 0 \le \phi \le \frac{\pi}{4} \qquad 0 \le \rho \le \cos(\phi)$



$$\iiint_E 1\,dV = \int_0^{\frac{\pi}{4}}\left[\int_0^{2\pi}\left[\int_0^{\cos(\phi)} 1\cdot\rho^2\sin(\phi)\,d\rho\right]d\theta\right]d\phi$$ not seperable

$$= \int_0^{\frac{\pi}{4}}\left[\int_0^{2\pi}\left[\frac{\rho^3\sin(\phi)}{3}\right]_{\rho=0}^{\rho=\cos(\phi)} d\theta\right]d\phi$$

$$= \int_0^{\frac{\pi}{4}}\left[\int_0^{2\pi}\frac{\cos^3(\phi)\sin(\phi)}{3}\,d\theta\right]d\phi$$ seperablo

$$= \left(\int_0^{\frac{\pi}{4}}\frac{\cos^3(\phi)\sin(\phi)}{3}\,d\phi\right)\left(\int_0^{2\pi} 1\,d\theta\right)$$

use substitution again

$$= \left[-\frac{1}{12}\cos^4(\phi)\right]_0^{\frac{\pi}{4}}\left[\theta\right]_0^{2\pi}$$

$$= \frac{1}{16}\cdot 2\pi$$

$$= \frac{\pi}{8}$$